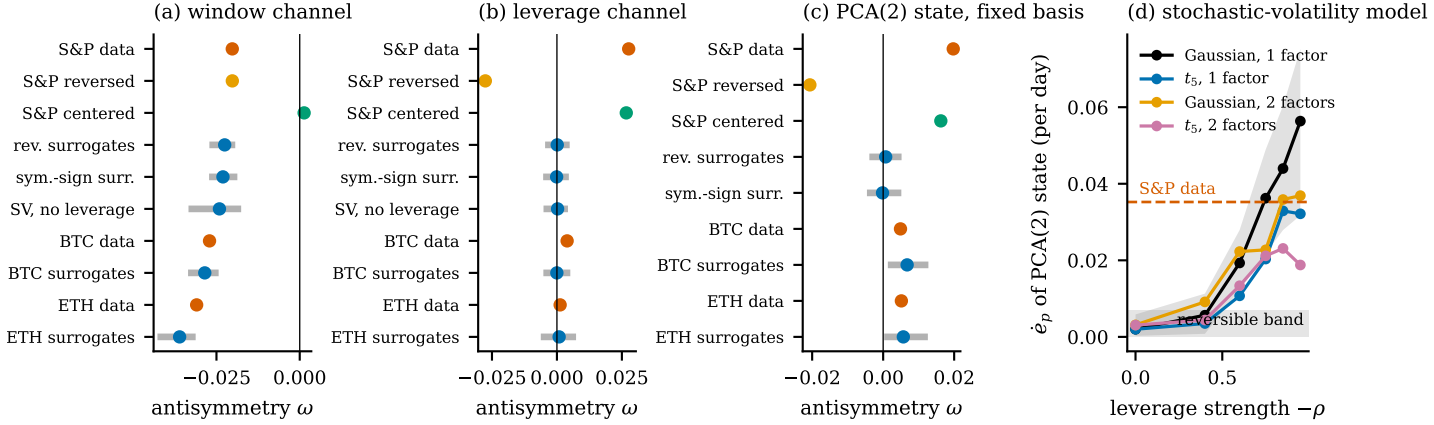
(a) window channel
S&P data
S&P reversed
S&P centered
rev. surrogates
sym.-sign surr.
SV, no leverage
BTC data
BTC surrogates
ETH data
ETH surrogates
−0.025
0.000
antisymmetry ω
(b) leverage channel
S&P data
S&P reversed
S&P centered
rev. surrogates
sym.-sign surr.
SV, no leverage
BTC data
BTC surrogates
ETH data
ETH surrogates
−0.025
0.000
0.025
antisymmetry ω
(c) PCA(2) state, fixed basis
S&P data
S&P reversed
S&P centered
rev. surrogates
sym.-sign surr.
BTC data
BTC surrogates
ETH data
ETH surrogates
−0.02
0.00
0.02
antisymmetry ω
(d) stochastic-volatility model
Gaussian, 1 factor
t_5, 1 factor
Gaussian, 2 factors
t_5, 2 factors
S&P data
reversible band
$\dot{e}_p$ of PCA(2) state (per day)
0.00
0.02
0.04
0.06
0.0
0.5
leverage strength $-\rho$